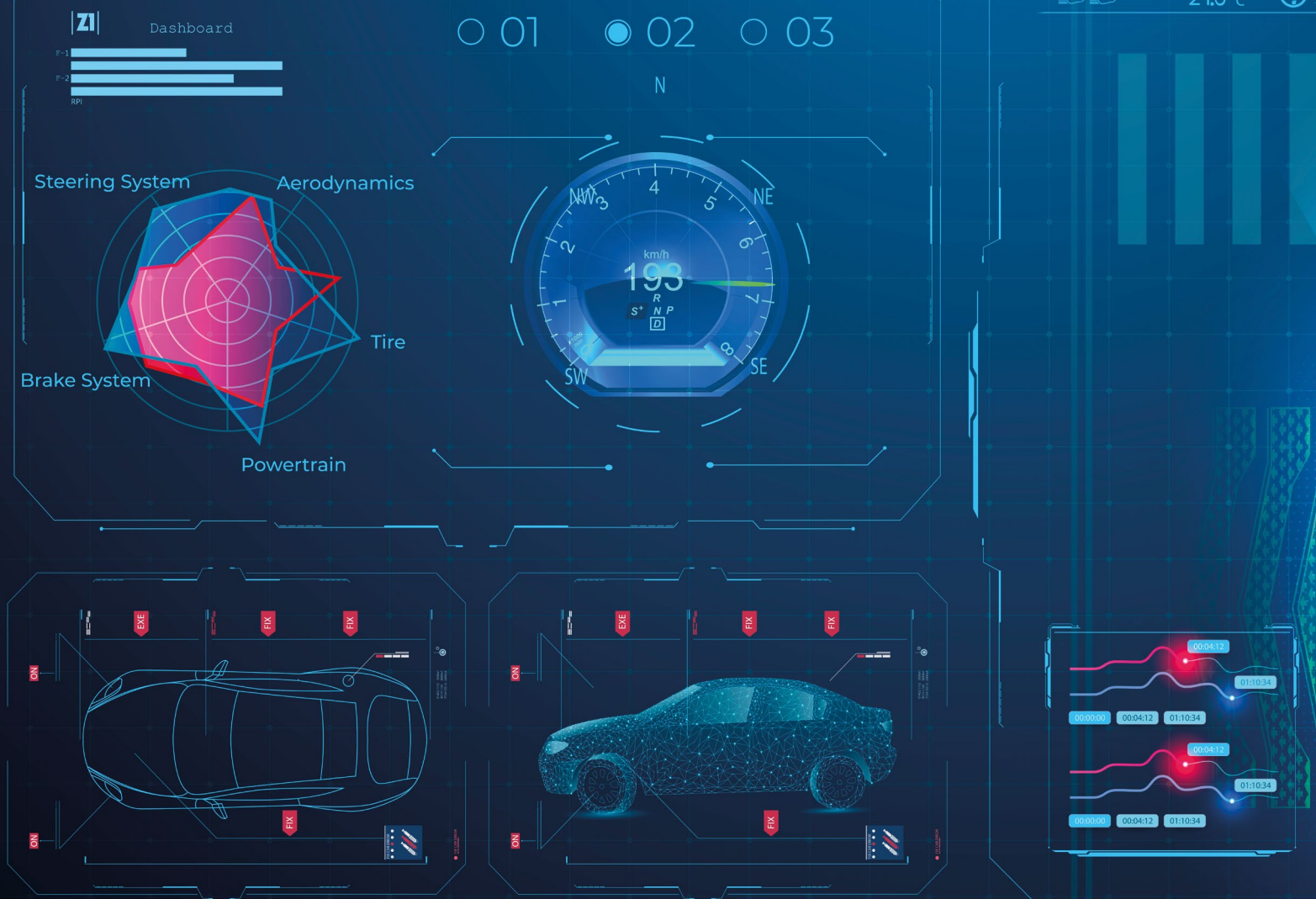




Data and Workflow Management for Virtual Development of Software-defined Vehicles

Today, the major changes in the automotive industry no longer take place in the classic areas of vehicle engineering or mechanical engineering, but primarily in the realm of software and the related control units. IPG Automotive presents by means of an example how the workflow in vehicle development can be designed by means of a newly developed tool environment in combination with new development approaches.



Virtual test driving has evolved into an indispensable basis for design and development in the automotive industry over the last decades. Originally used for tests in the area of classic vehicle dynamics, this test method has been subjected to continuous further development based on changing challenges.

For reasons of scalability and cost efficiency, attempts are currently being made to virtualize as many areas of testing as possible, **FIGURE 1**. Whereas individual software modules are frequently subjected to isolated testing in simulation, their integration often takes place only in the vehicle.

Since simulation, along with the vehicles themselves, is becoming increasingly complex, more and more specializations have emerged for specific realms of simulation compared to the past when one individual used to be solely responsible for running an entire simulation effort. Additional difficulties arise from the fact that virtual development processes cannot be integrated into existing structures, and that user trust in simulation results is often lacking. This article presents an exemplary virtual development method with which these challenges can be met and that additionally enables efficient development processes in terms of costs and time.

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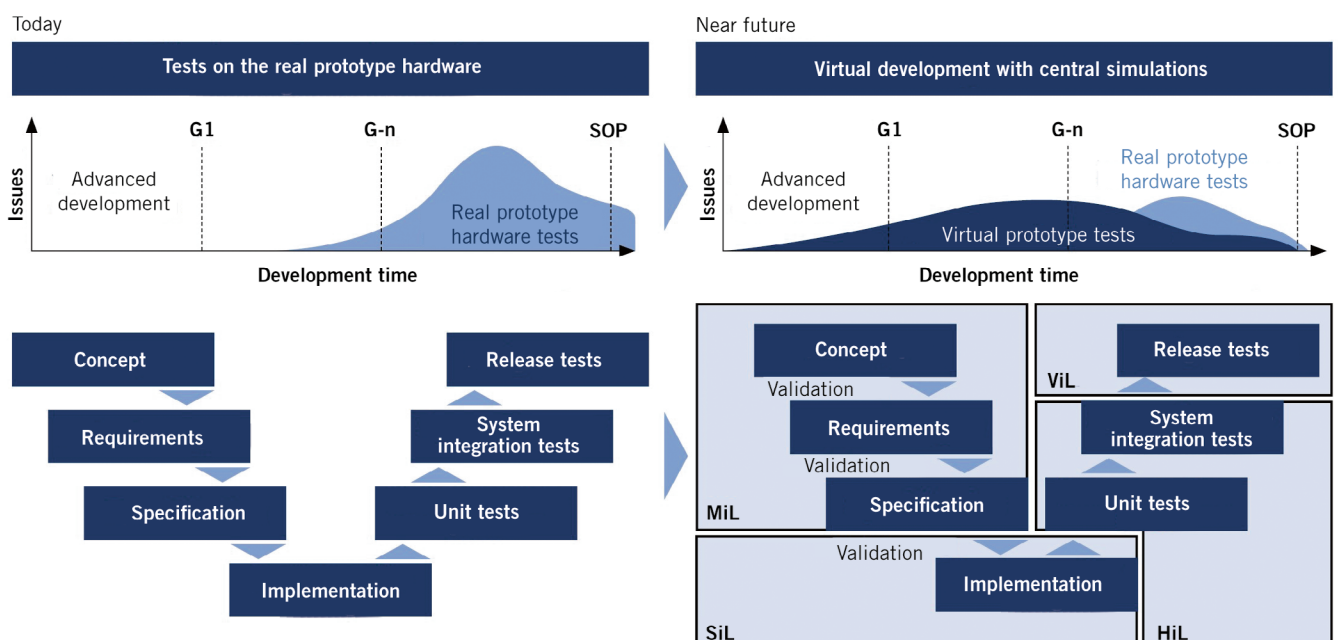


FIGURE 1 Difference between test driving based on real-world prototypes and virtual vehicle development including central simulation management (G: gate, SOP: start of production) (© IPG Automotive)

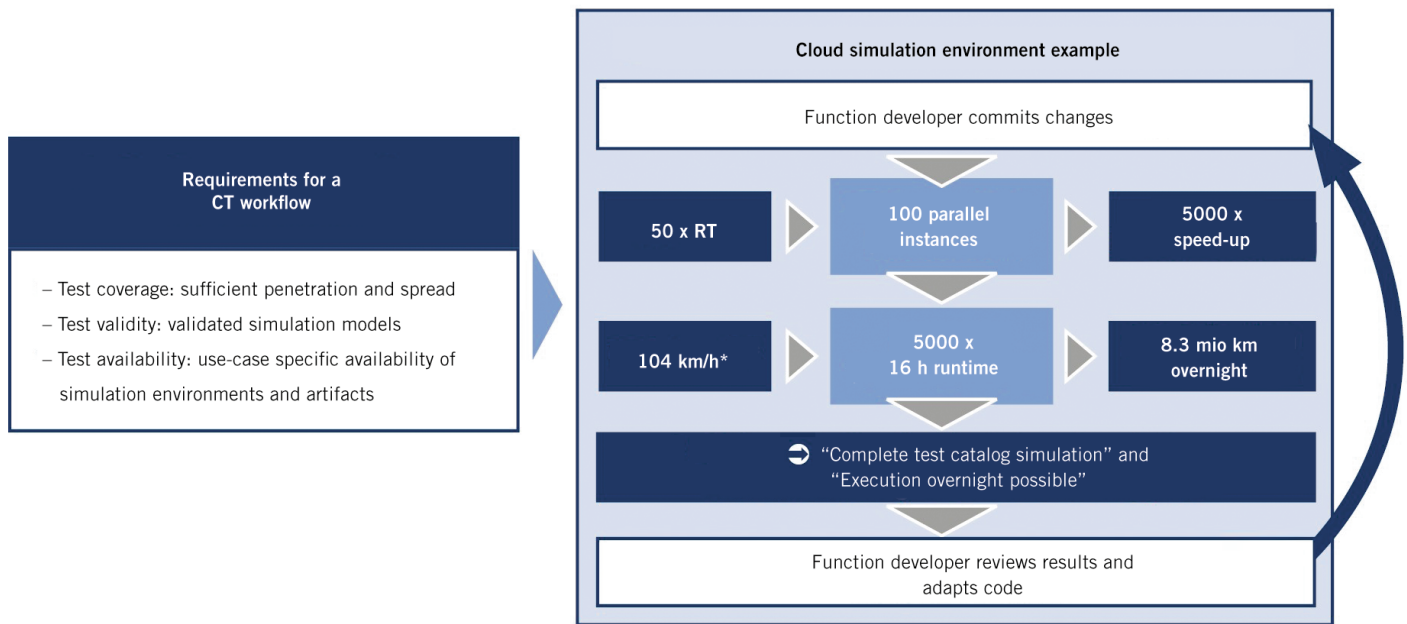


FIGURE 2 Requirements and processing times of an automated test pipeline (© IPG Automotive)

*Average speed on German motorways 2022

INCREASING VEHICLE COMPLEXITY

The complexity of modern vehicles is increasing primarily for one reason: their software content keeps growing. While vehicles used to differ from each other mainly due to hardware components, software is now becoming the central differentiator and key factor of vehicle characteristics. This evolution is referred to as the software-defined vehicle [1-3]. Increasingly extensive software functions with hundreds of millions of lines of code and numerous interdependencies confront vehicle manufacturers with major challenges in terms of testing requirements.

This leads to a clearly growing complexity of future vehicles that in turn poses new challenges regarding their connectivity. At the same time, the pressure to heavily reduce development cycles keeps growing. This, for instance, is caused by end customers increasingly expecting Over-the-Air (OTA) software updates – enabling improvements and new functionalities for vehicles even after several years in the field – to be provided across the entire product lifecycle. As a result, every vehicle becomes one-of-a-kind and keeps changing, which entails high costs and efforts for validation and approval of the development.

Highly automated driving functions considerably add to those challenges because, due to the multitude of variable parameters, skyrocketing number of required test kilometers and scenarios, as well as limited reproducibility, they cannot be tested and validated exclusively in real-world road testing [4]. Especially functional safety, defined in ISO 26262, that is required constitutes a central aspect of vehicle development [5]. This proof of safety states that manufacturers of safety-related systems in the automotive industry must develop the systems according to the current state of science and technology in order to comply with their legal duty to maintain safety and to ensure the intended function of the systems. To provide such proof – in line with the automotive systems engineering approach – the vehicle must always be viewed as a total system including all sensors, actuators and electronic control units involved. This approach encompasses the development and validation of the systems and driving functions in the total vehicle considering realistic scenarios and leads to a significant increase in efficiency and traceability [4, 6-8].

Virtual prototypes undergoing an evolution during their development and lifecycle play a key role in this: they may represent replicas of real-world vehicles validated according to specific

use cases and include models of the relevant (sub-)systems of the real-world vehicles on which they are modeled. In that way, they provide the basis for agile development and validation between the defined quality gates as well as for management and approval of OTA updates during the lifecycle in the field. Because, due to the short intervals between updates, a validation of new software versions based on real-world vehicles is feasible only to a limited extent, the unlimited availability of virtual prototypes provides an enormous benefit. To enable this agile development method, comprehensive management of the required data and parameterizations as well as of the software components for the virtual prototypes is indispensable [9].

POSSIBLE APPROACHES

Because the importance of software in automotive development keeps increasing, assurance of its quality is one of the central challenges. However, for the various levels of road approval, for example, of new driving functions, tests in the total vehicle are currently mandatory. In reality, that is nearly impossible and no longer economically feasible due to the number of necessary tests and increasingly shorter development cycles.

Established software development methods such as Continuous Integration, Testing and Deployment (CI/CT/CD) can also be used in the automotive environment to develop new workflows. They help meet the aforementioned challenges in the entire product lifecycle. When using this development method, the software components are continuously created, tested and improved, for instance by means of daily builds. The individual process steps can be largely automated. When adjusting a software module, the total system is automatically composed from the current versions of the individual components. Following the successful completion of this build process, module and integration tests are run on software level (CI/CT).

Traditionally, the software created in this way would be integrated into a test vehicle and tested in a real-world road test. However, depending on the availability of the test vehicle, this can normally be done only several days following the integration of the software, which constitutes a disruption of the closed CI/CT chain. By contrast, Virtual Prototypes (VP) enable the seamless continuity of this chain. If the software build process is extended by the VP build process it enables the seamless continuation of automated testing in virtual test driving. Due to parallel execution of the tests in the

cloud, for example, an enormously large number of test kilometers can be virtually driven per day. This enables high coverage in terms of testing penetration and spread. For virtual tests in freeway scenarios, approximately 8 mio test kilometers per night are achieved in that way. As a result, a clearly higher quality level of the software that is to be developed (an individual driving function or a complete AD driving stack) is attained in a much shorter period of time, **FIGURE 2**.

The constant availability of all required data – such as data for the virtual prototypes – is a basic prerequisite. The ability to build them correctly presupposes the presence of component data for parameterizing the simulation models. In addition, data pertaining to the functional software in the vehicle, including calibration data, data for describing scenarios, as well as measured data from sensors and about driving mode, are required. A facility for storing and comparing the data of simulation results and potential test data, and for deriving the necessary analysis for enhancing the software from them, is required as well.

To enable this complexity to be managed, a superordinate level for the simulation is needed. That level must provide the possibility to manage and revise data volumes and to easily make them available to users and use cases. An exem-

plary description of a possible development and test environment for this purpose is provided in the following.

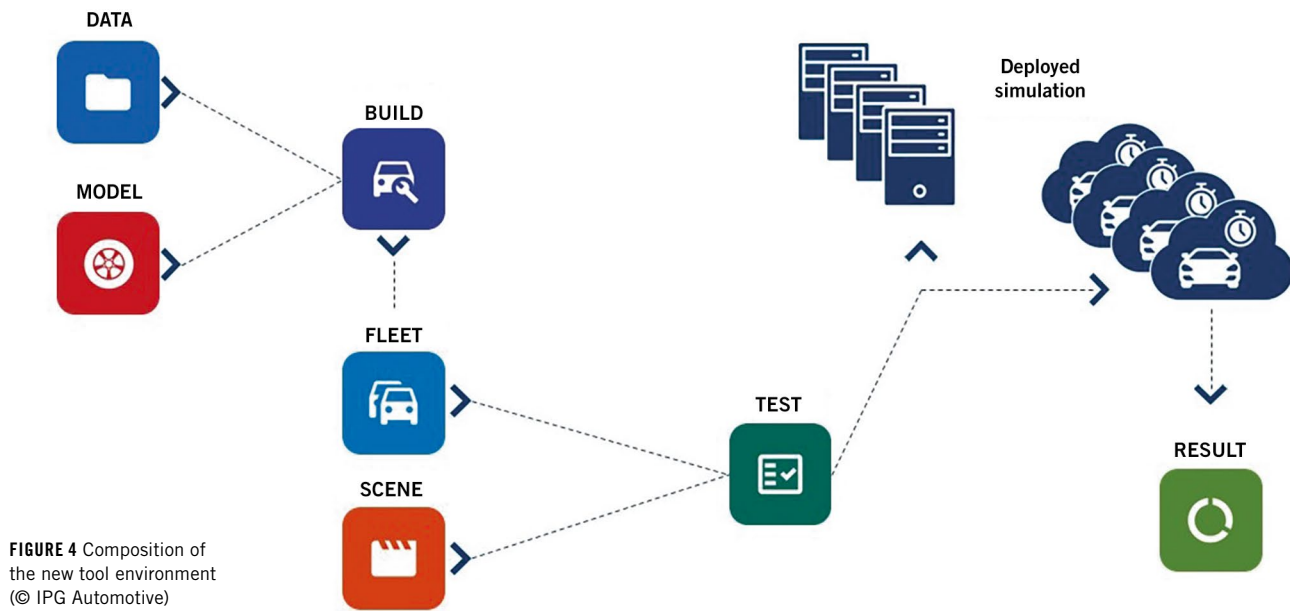
NEW TOOL ENVIRONMENT

Such an environment has to meet various prerequisites. A fundamental one is that it makes a large number of test platforms (MiL/SiL/HiL/ViL, open-/closed-loop tests, proving ground tests with real targets, etc.) centrally available to all development teams – across departmental and company boundaries. Agile software development processes benefit from maximum virtualization, high scalability and fast test cycles. The simulation serves as a means to an end in this regard and is to be understood purely as a tool. The objective is to obtain the required test results. However, in pursuit of that objective, the development team's need to familiarize itself with new (simulation) topics should be minimized so that the team can dedicate its time to its actual development work. Even developers without any simulation experience, such as pure software developers, should be able to use simulation – in the best case without even being aware of it, for instance, when the simulation is automatically started in the background.

To ensure that and to establish natural workflows, IPG Automotive devel-

FIGURE 3 Exemplary representation of the recurring software integration workflow (© IPG Automotive)





oped a new tool environment. It represents a collection of individual tools that can be used either independently of each other or as a full toolkit. It addresses users (OEMs, suppliers, or software companies) who have already deployed simulation in their development process and who, through meaningful data and workflow management, can achieve better software quality in less time, and experience an efficiency increase, **FIGURE 3**.

The basis of this environment is comprehensive data management enabling revisioning and linking of data. Taking intellectual property into account, these data can be shared across company boundaries. The provision of services for data management, simulation performance and simulation results analysis as well as possibilities for comparison (for data and simulation results) considerably facilitate workflows.

Users from a wide variety of functions can work together seamlessly in this shared development environment. The creation of valid virtual prototypes is equally covered as the management of complex simulation scenarios. The environment is designed for seamless adaptation to existing processes at OEMs and suppliers. The resulting toolkit offers users the opportunity to optimize their individual development in the necessary places. The tool environment can also be deployed on component level and thus

integrates itself with existing tool landscapes as needed.

The central infrastructure enables cross-domain utilization and centralized management of the fleet of virtual prototypes, test scenarios and test results, **FIGURE 4**. The enabled work/task distribution and related opportunity for fast and effective collaboration enhances efficiencies in the development and approval process especially of software components. Data quality and maturity across the entire lifecycle are ensured by versioning and traceability.

SUMMARY AND OUTLOOK

The increase in software-defined vehicles results in new requirements pertaining to their development and operation. Superordinate data and workflow management enabling efficient development and testing work is becoming increasingly important, and simulation as an individual discipline continues to be of central importance.

In the coming years, further developments in this area are anticipated. For instance, software will increasingly have to be approved for vehicles without the opportunity to extensively test it in the real-world vehicle. Likewise, vehicle functions that the consumer can acquire for retrofitting are becoming more and more relevant for vehicles in the field. Testing of such functions

is also feasible by means of simulating a digital twin prior to OTA delivery. For efficient execution, massive parallelization of virtual tests is indispensable to ensuring software integrity and thus vehicle safety following OTA updates.

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